25901128 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBOTA07 - PRINCIPLES OF AGRONOMY AND HORTICULTURE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very short answer

Answer any **10** questions

Each question carries **2** marks

1. Explain horticulture. [U] / [CO1]
2. Define secondary tillage. [U] / [CO2]
3. Define layering. [U] / [CO3]
4. Write the recommended age of solanaceous seedlings for transplanting. [K] / [CO4]
5. Define nutrient deficiency in plants. [K] / [CO2]
6. What is meant by shade house nursery? [K] / [CO3]
7. Name any two components of INM. [K] / [CO2]
8. What is urea? [U] / [CO2]
9. Abiu belongs to which family? [K] / [CO4]
10. What is meant by indigenous agricultural practices? [K] / [CO1]

11. Explain the objectives of White Revolution. [K] / [CO1]

12. What is meant by laterite soil? [U] / [CO2]

[2x10 = 20]

Part B

Short Essay

Answer any **8** questions

Each question carries **5** marks

13. Discuss the use of shrubs and trees in garden planning and landscaping. [U] / [CO3]

14. Explain how biological agents help in sustainable pest management. [U] / [CO4]

15. Describe the procedure for preparing and planting cuttings. [U] / [CO3]

16. Write a note on the modern developments in Indian agriculture. [U] / [CO1]

17. Explain the process and benefits of vermicomposting. [U] / [CO2]

18. Explain the advantages and disadvantages of budding. [U] / [CO3]

19. List and describe the sequential steps involved in the process of hybridization. [U] / [CO1, CO4]

20. Describe the current trends in agricultural policies and practices in India. [K] / [CO1]

21. Explain the role of agronomy in sustainable crop production. [K] / [CO1]

22. Discuss the factors affecting soil fertility. [U] / [CO2]

[5x8 = 40]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

23. Describe the different types of grafting and their applications in horticulture. [U] / [CO3]

24. Give an essay on disease and pest management in Mango. [A] / [CO4]

[10x1 = 10]